HTTP Guide for Thaghrah

Description

HTTP (Hypertext Transfer Protocol) is the foundational protocol used for communication on the World Wide Web. It defines how clients (such as web browsers) and servers exchange information in a standardized way.

HTTP follows a **client-server, request-response** model. The client sends a request to the server, and the server returns a response. It is a **stateless** protocol — each request is independent, and the server does not retain memory of previous requests unless additional mechanisms (like cookies) are used.

HTTP is **text-based**, making it human-readable, and it typically operates on top of TCP. The most common version in use is HTTP/1.1, though modern implementations often use HTTP/2 or HTTP/3 for better performance.

Key Components of HTTP

1. HTTP Requests

An HTTP request consists of:

- **Request Line**: Method + URI (path) + HTTP Version Example: GET /index.html HTTP/1.1
- **Headers**: Metadata about the request (e.g., Host, User-Agent, Accept, Cookie)
- **Body** (optional): Data sent with methods like POST or PUT

Common HTTP Methods:

- **GET**: Retrieve data from the server
- **POST**: Send data to the server (e.g., form submissions)
- **PUT**: Replace or create a resource
- **DELETE**: Remove a resource
- **HEAD**: Retrieve headers only (no body)

2. HTTP Responses

An HTTP response consists of:

- **Status Line**: HTTP Version + Status Code + Reason Phrase Example: HTTP/1.1 200 OK
- **Headers**: Metadata about the response (e.g., Content-Type, Set-Cookie, Location)
- **Body** (optional): The actual content (HTML, JSON, images, etc.)

3. Status Code Categories

Status codes are grouped into five classes based on the first digit:

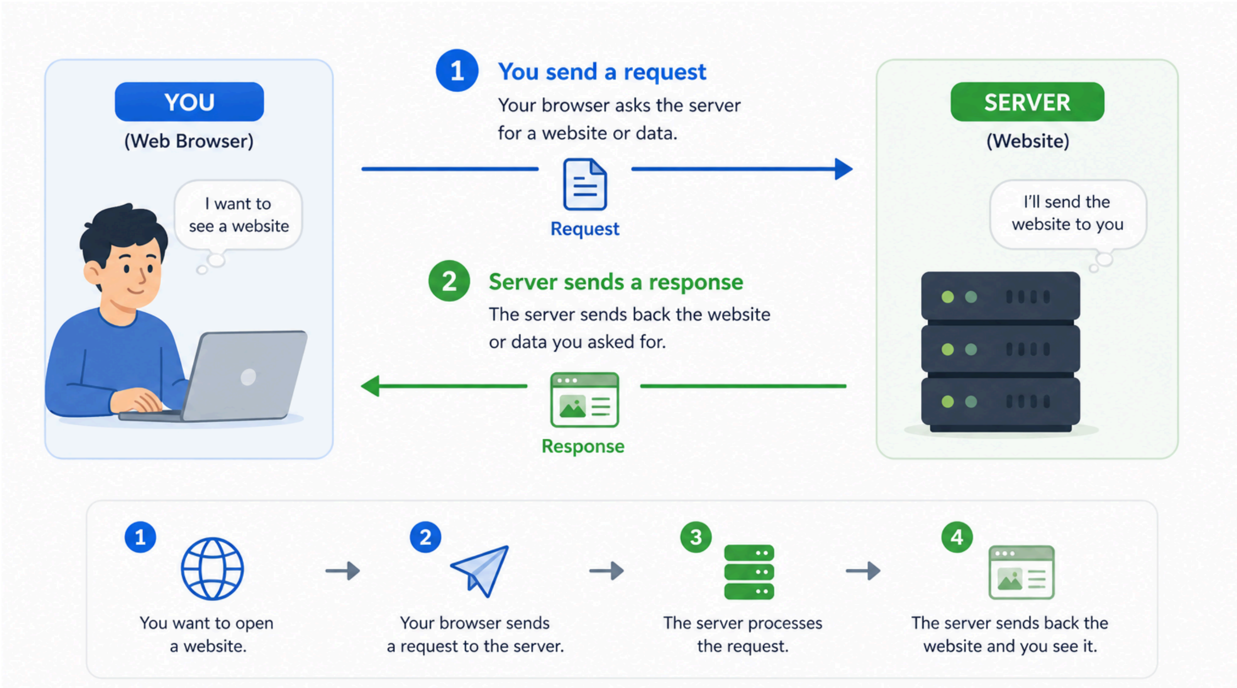
Class	Meaning	Example Codes
1xx	Informational	100 Continue
2xx	Success	200 OK
3xx	Redirection	301 Moved Permanently, 302 Found
4xx	Client Error	404 Not Found
5xx	Server Error	500 Internal Server Error

4. Important Features

- **Cookies**: Used to maintain state (session management) across stateless requests. Set by the server via the Set-Cookie header and sent back by the client via the Cookie header.
- **Redirects**: Server instructs the client to go to another URL using 3xx status codes and the Location header.
- **Content Negotiation**: Headers like Accept and Content-Type allow clients and servers to agree on data formats.

Visualizations

HTTP Request-Response Cycle:



HTTP Explained in 3 Minutes

https://www.youtube.com/watch?v=KvGi-UDfy00